

The Exponent Two Is Not Optimal for a Two-Dimensional Critical Hardy Stability Inequality

Xiaojian Zeng
PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

On the unit disk in two dimensions, consider the critical Hardy deficit with its logarithmic weight and the Luxemburg-distance from the corresponding virtual extremals. A previously established stability estimate has distance exponent two. We prove that every exponent $q > 1$ is admissible, with an explicit choice of the exponential parameter. Consequently the threshold satisfies $q_* \leq 1$, and in particular it is not equal to two. The argument combines a ground-state-transform L^2 remainder with a subcritical exponential interpolation inequality. It does not determine the endpoint $q = 1$ or the exact value of q_* .

1 Statement and context

Let $B_1(0) \subset \mathbb{R}^2$, put

$$t(r) = \log \frac{2}{r}, \quad t_0 = \log 2, \quad \omega(x) = \sqrt{t(|x|)},$$

and define

$$\mathcal{H}(u) = \int_{B_1(0)} |\nabla u|^2 dx - \frac{1}{4} \int_{B_1(0)} \frac{|u|^2}{|x|^2 t(|x|)^2} dx.$$

The Luxemburg norm used below is

$$\|u\|_{\text{Exp } L^2} := \inf \left\{ \lambda > 0 : \int_{B_1(0)} \left(e^{|u|^2/\lambda^2} - 1 \right) dx \leq 1 \right\}.$$

For $C > 0$ and nonzero $u \in W_0^{1,2}(B_1(0))$, set

$$D_C(u) = \inf_{a \in \mathbb{R}} \int_{B_1(0)} \left[\exp \left(\frac{C|u - a\omega|^2}{\|u\|_{\text{Exp } L^2}^2} \right) - 1 \right] dx.$$

Call $q > 0$ admissible if there are $c, C > 0$, independent of u , such that

$$\mathcal{H}(u) \geq c \|u\|_{\text{Exp } L^2}^2 D_C(u)^q \quad \text{for every } 0 \neq u \in W_0^{1,2}(B_1(0)), \quad (1.1)$$

and let q_* be the infimum of all admissible q .

Sahu's general-dimensional stability theorem gives exponent N for this Luxemburg-distance; in the present specialization it gives exponent two [2, Theorem 1.1]. Its accompanying remark notes that optimality of the exponent N in the general setting is not settled by that argument [2, Remark 1.1]. The following disk-specific estimate strictly improves the resulting upper bound on the threshold.

For broad adjacent context on sharp-constant methodology in a different, operator-theoretic setting, see [3]. That work is not used in, and does not support, any Hardy–Orlicz assertion made here.

Theorem 1. *For every $q > 1$, define*

$$C_q = \frac{q-1}{2q}, \quad p_q = \frac{q+1}{2},$$

and

$$K = \int_0^1 r t(r) \log \frac{t(r)}{t_0} dr.$$

Then $0 < K < \infty$, and every nonzero $u \in W_0^{1,2}(B_1(0))$ satisfies

$$\mathcal{H}(u) \geq c_q \|u\|_{\text{Exp } L^2}^2 D_{C_q}(u)^q, \quad c_q = \frac{1}{\max\{1, e^q K^{p_q}\}} > 0. \quad (1.2)$$

Consequently $(1, \infty)$ consists of admissible exponents, so $q_ \leq 1$ and the equality $q_* = 2$ is false.*

2 An L^2 remainder for the Hardy deficit

Lemma 2. *For every $u \in W_0^{1,2}(B_1(0))$,*

$$\|u\|_{L^2(B_1(0))}^2 \leq K \mathcal{H}(u), \quad (2.1)$$

where

$$K = 4 \int_{t_0}^{\infty} t \log \frac{t}{t_0} e^{-2t} dt < \infty. \quad (2.2)$$

Proof. We first take $u \in C_c^\infty(B_1(0))$ and write $\varphi = u/\omega$. Direct differentiation gives

$$\omega'(r) = -\frac{1}{2r\sqrt{t(r)}}, \quad -\Delta\omega = \frac{\omega}{4r^2 t(r)^2}.$$

Integration by parts on $\{\varepsilon < |x| < 1\}$, followed by $\varepsilon \downarrow 0$, therefore yields the ground-state identity

$$\mathcal{H}(u) = \int_{B_1(0)} \omega^2 |\nabla \varphi|^2 dx. \quad (2.3)$$

Indeed, the outer boundary term vanishes, while the inner one has absolute value

$$\int_0^{2\pi} \frac{|u(\varepsilon, \vartheta)|^2}{2t(\varepsilon)} d\vartheta,$$

which tends to zero.

For each polar angle, the zero boundary value and Cauchy–Schwarz give

$$\begin{aligned} |\varphi(r, \vartheta)|^2 &\leq \left(\int_r^1 st(s) |\partial_s \varphi(s, \vartheta)|^2 ds \right) \left(\int_r^1 \frac{ds}{st(s)} \right) \\ &\leq E_\vartheta \log \frac{t(r)}{t_0}, \end{aligned}$$

where

$$E_{\vartheta} = \int_0^1 st(s) |\partial_s \varphi(s, \vartheta)|^2 ds.$$

Multiplying by $rt(r)$ and integrating in r and ϑ gives

$$\|u\|_2^2 \leq K \int_0^{2\pi} E_{\vartheta} d\vartheta \leq K\mathcal{H}(u),$$

because the angular contribution in (2.3) is nonnegative. The change of variables $t = \log(2/r)$ gives (2.2), and exponential decay proves that K is finite.

It remains only to pass to the stated Sobolev space. Identity (2.3) first implies, for test functions v ,

$$\int_{B_1(0)} \frac{|v|^2}{|x|^2 t(|x|)^2} dx \leq 4 \int_{B_1(0)} |\nabla v|^2 dx. \quad (2.4)$$

Thus approximation in $W_0^{1,2}(B_1(0))$ also gives convergence in the weighted L^2 norm in (2.4). Both terms defining $\mathcal{H}$, as well as the ordinary L^2 norm, consequently pass to the limit, proving (2.1). $\square$

3 Exponential interpolation

Lemma 3. *If $0 < \theta < 1$ and $x \geq 0$, then*

$$e^{\theta x} - 1 \leq e x^{1-\theta} (e^x - 1)^{\theta}. \quad (3.1)$$

Consequently, for every nonnegative measurable X for which the right-hand side below is finite,

$$\int_{B_1(0)} (e^{\theta X} - 1) dx \leq e \left(\int_{B_1(0)} X dx \right)^{1-\theta} \left(\int_{B_1(0)} (e^X - 1) dx \right)^{\theta}. \quad (3.2)$$

Proof. For $0 \leq x \leq 1$, one has $e^{\theta x} - 1 \leq ex$, while $x^{1-\theta}(e^x - 1)^{\theta} \geq x$. For $x \geq 1$,

$$e^{\theta x} - 1 \leq e^{\theta x} \quad \text{and} \quad e x^{1-\theta} (e^x - 1)^{\theta} \geq e(1 - e^{-1})^{\theta} e^{\theta x} > e^{\theta x}.$$

This proves (3.1); Hölder's inequality gives (3.2). $\square$

4 Proof of the threshold bound

Proof of Theorem 1. Fix $0 \neq u \in W_0^{1,2}(B_1(0))$ and write

$$N = \|u\|_{\text{Exp } L^2}, \quad X = \frac{|u|^2}{N^2}, \quad h = \frac{\mathcal{H}(u)}{N^2}.$$

The Trudinger–Moser embedding ensures $0 < N < \infty$ [1]. Taking admissible Luxemburg parameters decreasing to the infimum and using monotone convergence gives

$$\int_{B_1(0)} (e^X - 1) dx \leq 1. \quad (4.1)$$

For $0 < \theta < 1$, choose the permitted competitor $a = 0$ in the definition of D_{θ} . Lemmas 2 and 3, together with (4.1), yield

$$D_{\theta}(u) \leq \min \left\{ 1, e(Kh)^{1-\theta} \right\}. \quad (4.2)$$

The first bound follows directly from $e^{\theta X} - 1 \leq e^X - 1$; the second uses $\int_{B_1(0)} X \, dx \leq Kh$.

Now fix $q > 1$ and put $\theta = C_q$. Then

$$q(1 - \theta) = \frac{q+1}{2} = p_q > 1.$$

If $h \leq 1$, equation (4.2) gives

$$D_{C_q}(u)^q \leq e^q K^{p_q} h^{p_q} \leq e^q K^{p_q} h.$$

If $h \geq 1$, the other part of (4.2) gives $D_{C_q}(u)^q \leq 1 \leq h$. Hence

$$h \geq \frac{D_{C_q}(u)^q}{\max\{1, e^q K^{p_q}\}}.$$

Multiplication by N^2 proves (1.2). Since this holds for every $q > 1$, taking the infimum of the admissible exponents proves $q_* \leq 1$. $\square$

Scope and assistance disclosure

The proof establishes only the upper bound $q_* \leq 1$. It makes no assertion about admissibility at $q = 1$, any lower bound for q_* , or the exact value of the threshold. No numerical experiment or computer-algebra calculation is used. An AI language model assisted with proof exploration, checks of the displayed algebra and quantifiers, and preparation of the manuscript. The author is responsible for the statements and proof presented here.

References

- [1] J. Moser. A sharp form of an inequality by N. Trudinger. *Indiana University Mathematics Journal*, 20(11):1077–1092, 1971.
- [2] Vivek Sahu. Improved quantitative stability for the critical Hardy inequality, 2026. arXiv:2608.19732 [math.AP].
- [3] Zijian Zeng. Fixed convex-lens spectral constants: Möbius reduction, sharp model theorems, and angle-dependent bounds. *Preprints*, August 2026.